



Handbook of Materials Failure Analysis with Case Studies from the Oil and Gas Industry

2016, Pages 353-370

Chapter 17 - Failure of 17-4PH stainless steel components in offshore platforms

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<https://doi.org/10.1016/B978-0-08-100117-2.00019-4> ↗

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Abstract

Precipitation hardening is a strengthening mechanism very common in many classes of metallic materials, from Al and Cu alloys to high-strength steels. In special, precipitation hardening (PH) stainless steels may allow corrosion resistance and mechanical strength desired to special applications. Despite this, there are many cases of failures involving PH stainless steels reported in the literature. Incorrect material selection or wrong heat treatment procedures are usually associated to these cases of failure. This chapter is focused on three cases of failure of 17-4PH steel, a common precipitation hardenable martensitic stainless steel used in oil and gas production and refining industries.

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2020, Engineering Failure Analysis

Citation Excerpt :

...High-strength steels are more susceptible to environmentally assisted cracking. Previous studies showed that inappropriate heat treatment may cause hydrogen embrittlement (HE) and stress corrosion cracking (SCC) [14,15]. Tavares et al. [16] investigated two failures of springs on pressure safety valve (PSV)....

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2019, Additive Manufacturing

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...The solution treatment and cooling of material to room temperature results in a martensitic structure but it yet lacks the high hardness. Aging the 17-4 pH results in precipitation hardening due to the formation of a rich copper precipitates [10]. The use of a double wire to achieve higher deposition rate WAAM was reported in [7], where one and two wires were fed into a plasma-based WAAM setup, for austenitic ferritic stainless steel....

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